

Residual-Adaptive Column Generation for Convex Two-Layer ReLU Fitting

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Abstract

We consider fitting a scalar-output two-layer ReLU network with weight decay, which is equivalent to a convex regularized fitting problem whose dictionary contains one candidate feature for every direction on the unit sphere. The convex problem is easy once the directions are fixed—it is an ordinary lasso—but selecting the best missing direction is NP-hard. We propose residual-adaptive column generation (RCG), which maintains a small working set of directions, refits their coefficients, and searches for new directions correlated with the current residual. When the restricted problems and the search are solved exactly, the objective error after t rounds is at most $2\tau^2 R^2/(t+4)$, where τ bounds the mass (coefficient sum) of an optimum and R bounds the atom norms; the bound carries no factor of the exponential number of activation patterns, although the method need not terminate finitely. Semidefinite relaxations give upper bounds on the search value, and hence dual bounds, in exact arithmetic; our floating-point evaluation of these bounds is a numerical estimate, not a certificate. In exploratory experiments the implementation matches strong reference values on planted instances, maintains its accuracy at a fixed working-set budget as random sampling degrades with dimension, and, on three real datasets against tuned Adam, AdamW, and SGD at budget-matched wall clock, reaches an objective 45% below the best tuned run on one dataset, while on the other two the best tuned runs finish 3.4% and 5.5% lower; these comparisons are exploratory, under budget-matched tuning grids. A controlled benchmark of implementation choices identifies pricing search and working-set consolidation as the dominant costs.

1 Introduction

Training a two-layer ReLU network is ordinarily a nonconvex problem, attacked by stochastic local search over learning rates, initializations, and restarts. For the scalar-output network with weight decay this is the problem

$$\text{minimize} \quad \frac{1}{2} \left\| \sum_{j=1}^m (Xw_j)_+ b_j - y \right\|_2^2 + \frac{\beta}{2} \sum_{j=1}^m (\|w_j\|_2^2 + b_j^2), \quad (1)$$

with variables the hidden weights $w_j \in \mathbf{R}^d$ and output weights $b_j \in \mathbf{R}$; the data matrix $X \in \mathbf{R}^{n \times d}$, responses $y \in \mathbf{R}^n$, width m , and decay parameter $\beta > 0$ are given, and $(z)_+ = \max(z, 0)$ acts entrywise. A remarkable line of work [13, 28] showed that this nonconvex problem is equivalent to a finite *convex* program, so that global optimality—and with it, an end to per-instance optimizer tuning—is at least meaningful.

The obstacle is size. The convex reformulation has one block of variables for every *activation pattern* the data admit, *i.e.*, every region of the hyperplane arrangement $\{x_i^T u = 0\}$ cut by the rows of X (Figure 1). With $r = \mathbf{rank} X$, the number of such regions can reach $2 \sum_{k=0}^{r-1} \binom{n-1}{k}$, so the program cannot be written down beyond small effective rank. Existing practice either samples a fixed random subset of patterns and solves the restricted program [21, 23], giving up any statement about the patterns not sampled, or returns to nonconvex local search, giving up the convex view entirely.

There is a third route. The convex program is an *atomic-norm* problem [7]: its dictionary is the continuum of ReLU features $(Xu)_+$ with $\|u\|_2 \leq 1$, and one never needs the whole dictionary, only a way to *price* it—to find the feature most correlated with a residual. Pricing over all exponentially many patterns collapses to a single continuous problem in u (Theorem 3), so column generation applies: keep a small working set of directions, solve the restricted problem (an ordinary lasso), and add the direction most correlated with the residual. We call this *residual-adaptive column generation* (RCG): every mechanism that grows or reshapes the working set is a function of the current residual.

Convexity has moved the hard part into an oracle; it has not removed it. Exact pricing is NP-hard (Theorem 12). This paper takes that oracle seriously in both directions. We prove exactly what an exact oracle buys: full optimality conditions and an $O(1/t)$ rate with no factor of the pattern count; we also prove that the method need not terminate finitely, even in two dimensions. And we quantify what a cheap oracle costs: pricing is NP-hard and APX-hard in general, polynomial at fixed effective rank, and admits computable semidefinite upper bounds in exact arithmetic.

The practical solver is built to one invariant: heuristics may only propose, and the objective disposes. Sketched multistart search, gated consolidation (§7), and semidefinite rounding generate candidate directions, but every acceptance is decided by the full-data objective of the original convex problem, so approximation degrades only the quality of the search, never the feasibility of the returned model or the objective being minimized. The solver’s gap diagnostics are numerical estimates, not certificates (Remark 17).

This paper. Our principal contribution is the exact analysis of the ideal method and the complexity landscape of its oracle: the pricing collapse and full Karush–Kuhn–Tucker (KKT) conditions with active-atom saturation (§3); the $2\tau^2 R^2/(t+4)$ rate from an empty working set, together with an explicit two-dimensional instance on which exact RCG runs forever (§4); and the hardness of pricing on rational inputs, already for the all-ones residual, complemented by fixed-rank tractability and a sampling-coverage bound (§5). A supporting contribution is a low-rank semidefinite dual (§6.2)—a $d \times d$ program rather than $n \times n$ —that upper-bounds the price in exact arithmetic and turns any valid bound into a dual lower bound on the fitting problem (§6). The empirical contribution is an objective-preserving solver (§7) and two grades of evidence for it (§8): exploratory comparisons on planted and

real data against tuned first-order baselines and random pattern sampling, and a controlled, preregistered benchmark of implementation choices with independent objective replay. Several individual results adapt known arguments, and we credit them where they appear; the hardness reduction, the nontermination construction, the small-cone identity, and the failure example for mixed-sign rounding are, to our knowledge, new.

1.1 Prior work

Convex ReLU formulations. Pilanci and Ergen [28] derive the finite convex reformulation of weight-decay training by enumerating activation patterns; Ergen and Pilanci [13] extend the equivalence, and the connection between weight decay and ℓ_1 -type regularization of the network’s function goes back at least to Neyshabur et al. [26], Savarese et al. [29]. The equivalence we state in §2 is a width-qualified version of these results. Our departure is to keep the dictionary continuous and search it, rather than enumerate or sample it.

Continuous dictionaries and conditional gradient. Incremental atom insertion for convex neural networks originates with Bengio et al. [3] and is developed for general continuous dictionaries by Bach [1]; the algorithmic template is Frank–Wolfe [8, 15, 20, 22], and alternating-descent conditional gradient adds local motion of active atoms [5, 12]. Our rate theorem is the penalized fully-corrective rate of this literature, proved directly for the signed-atomic objective, with antecedents in greedy approximation [2, 8]; the nontermination example appears to be new and is specific to the continuous ReLU dictionary.

Selected-pattern solvers. SCNN [23] and CRONOS [14] solve the convex program restricted to a sampled or selected set of patterns, with accelerated first-order, augmented-Lagrangian, and GPU-oriented methods; Kim and Pilanci [21] analyze what random sampling achieves. These are strong solvers of the restricted problem. Their residuals do not test the patterns they omit, which is precisely the separation step (§3) this paper studies; the random-pattern baseline in §8.2 isolates that difference while holding the rest of the method fixed.

Hardness. Training-time hardness for depth-2 ReLU networks is established by Goel et al. [17] and refined by parameterized results [16]; Bach [1] already notes the hardness of the incremental (oracle) step for related dictionaries, and Boob et al. [4] prove hardness of fitting a single ReLU, the problem closest to the pricing functional itself. Theorem 12 localizes hardness in the pricing functional itself, on rational inputs, already at the all-ones residual, and adds APX-hardness with a no-PTAS consequence via a rational Max-Cut reduction.

Semidefinite bounds. Our price bounds use the positive-semidefinite Boolean quadratic machinery of Nesterov [24] and Goemans and Williamson [18]. The diagonal dual we start from is classical—it is the Delorme–Poljak bound for Max-Cut [10]—and our contribution is the closed *small-cone* form, a $d \times d$ dual with an explicit recovery map that tolerates inexact inner solutions in exact arithmetic.

Outline. §2 sets up the three equivalent problems. §3 develops pricing, the optimality conditions, and the dual. §4 analyzes ideal exact RCG. §5 maps the complexity of pricing, and §6 constructs computable price bounds. §7 describes the solver, and §8 reports the numerical evidence. §9 concludes.

2 The convex fitting problem

Throughout, $X \in \mathbf{R}^{n \times d}$ has rows $x_1^T, \dots, x_n^T$, the responses are $y \in \mathbf{R}^n$, and $r = \mathbf{rank} X$ is the effective rank. For a direction $u \in \mathbf{R}^d$ with $\|u\|_2 \leq 1$, the *ReLU atom* $a(u) = (Xu)_+ \in \mathbf{R}^n$ is the feature vector the hidden neuron u contributes on the data. Define the compact signed atom set

$$\mathcal{A} = \{ \sigma (Xu)_+ : \|u\|_2 \leq 1, \sigma \in \{-1, 1\} \} \subset \mathbf{R}^n.$$

The finite-atom fitting problem is

$$\text{minimize } F(\alpha, U) = \frac{1}{2} \left\| \sum_{j=1}^P \alpha_j (Xu_j)_+ - y \right\|_2^2 + \lambda \sum_{j=1}^P |\alpha_j|, \quad (2)$$

with variables the number of atoms P (a nonnegative integer), the signed coefficients $\alpha \in \mathbf{R}^P$, and the directions u_j with $\|u_j\|_2 \leq 1$; the data X , y , and the penalty $\lambda > 0$ are given. The direction parameterization makes (2) nonconvex as written, but it is a finite representation of a convex problem. Let $\gamma_{\mathcal{A}}$ be the atomic gauge of $\mathcal{A}$,

$$\gamma_{\mathcal{A}}(\hat{y}) = \inf \left\{ \sum_j c_j \mid \hat{y} = \sum_j c_j a_j, c_j \geq 0, a_j \in \mathcal{A} \right\},$$

with value $+\infty$ outside the span of $\mathcal{A}$. We call $\sum_j c_j$ the *mass* of the representation. The convex problem is

$$\text{minimize } \frac{1}{2} \|\hat{y} - y\|_2^2 + \lambda \gamma_{\mathcal{A}}(\hat{y}), \quad (3)$$

with variable the prediction $\hat{y} \in \mathbf{R}^n$. Since $\mathcal{A}$ is compact and contains 0, its convex hull is compact, the gauge is closed, and the infimum in $\gamma_{\mathcal{A}}$ is attained; allowing arbitrary finite representations in (2) therefore gives exactly the optimal value of (3).

The weight-decay problem (1) is the same problem again, provided the network is wide enough.

Proposition 1 (Width-qualified equivalence). *Let $\lambda = \beta > 0$, and let m^* be the least number of signed atoms in a minimum-mass representation of some optimum of (3). If $m \geq m^*$, then problems (1), (2), and (3) have the same optimal value; every optimal atomic representation with at most m atoms yields an optimal network by the construction in the proof, and every optimal network yields an optimal atomic representation by normalization.*

Proof. For a nonzero neuron, put $u = w / \|w\|_2$ and $\alpha = \beta \|w\|_2$. Positive homogeneity of the ReLU preserves the neuron's prediction, while

$$\frac{\beta}{2} (\|w\|_2^2 + b^2) \geq \beta |b| \|w\|_2 = \lambda |\alpha|,$$

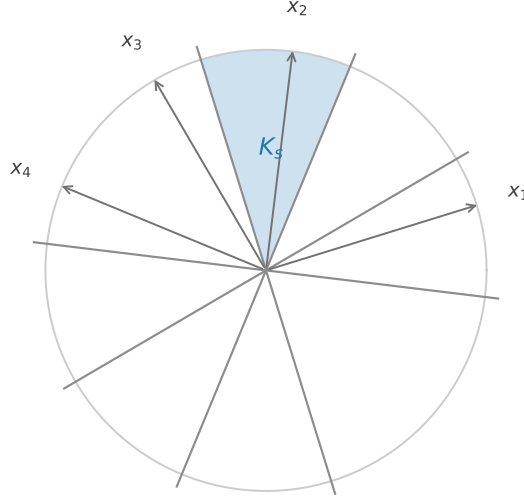


Figure 1: A central arrangement in $\mathbf{R}^2$ with $n = 4$ data rows. Each row x_i contributes the hyperplane $x_i^T u = 0$; the full-dimensional regions are the chambers, and the shaded cone is the chamber K_s on which all four rows are active, $s = (1, 1, 1, 1)$. Every ReLU atom is linear on each chamber.

with equality when the two layers are balanced, $\|w\|_2 = |b|$. Conversely, an atom with coefficient α is realized by the balanced neuron $w = \sqrt{|\alpha|} u$, $b = \mathbf{sign}(\alpha)\sqrt{|\alpha|}$, and the width hypothesis permits this construction atom by atom for an optimal representation. $\square$

Corollary 7 below shows $m \geq n$ is always sufficient. We do not claim the equivalence for an arbitrary smaller width.

Chambers. The finite convex program of Pilanci and Ergen [28] is a third representation of the same gauge, and it supplies the geometry we use throughout. Work in the row span of X , of dimension r . A *chamber* is a full-dimensional region of the central arrangement $\{x_i^T u = 0 : \|x_i\|_2 > 0\}$ restricted to the row span (the arrangement is *essential* there: the nonzero rows span it). For a binary gate vector $s \in \{0, 1\}^n$ write $D_s = \mathbf{diag}(s)$, where $\mathbf{diag}(v)$ is the diagonal matrix with diagonal v , and define the closed cone

$$K_s = \{u \in \mathbf{R}^d \mid (2D_s - I)Xu \geq 0\},$$

the set of directions whose sign pattern is consistent with s . The cone K_s lives in $\mathbf{R}^d$ and contains the null space of X ; its intersection with the row span is the corresponding chamber closure. Let $\mathcal{S}$ be the set of gate vectors of full-dimensional chambers. For $u \in K_s$ we have $D_s Xu = (Xu)_+$, including on the boundary, and the chamber program is

$$\text{minimize} \quad \frac{1}{2} \left\| \sum_{s \in \mathcal{S}} D_s X(v_s - w_s) - y \right\|_2^2 + \lambda \sum_{s \in \mathcal{S}} (\|v_s\|_2 + \|w_s\|_2), \quad (4)$$

with variables the chamber blocks $v_s, w_s \in K_s$. Normalizing a nonzero block, $v_s = \|v_s\|_2 u$ with $u \in K_s$, produces the signed atom $\|v_s\|_2 (Xu)_+$, so every feasible point of (4) is a representation in (3) of no larger cost; conversely, grouping the atoms of a representation by chamber and sign and using the triangle inequality on each group's directions (each K_s is convex) gives a chamber point of no larger cost. The two problems have the same value. Note one consequence: the restricted problem over any finite set of atoms needs *no* cone constraints—each ReLU atom is a valid feature of its own chamber automatically—so the restricted problem is an ordinary lasso.

Proposition 2 (Chamber count). *If $r \geq 1$, the number of full-dimensional chambers is at most*

$$2 \sum_{k=0}^{r-1} \binom{n-1}{k},$$

with equality when all n rows are nonzero and in general position in the row span (every subset of at most r of them linearly independent). If $r = 0$ there is one zero-dimensional chamber closure.

Proof. The bound is the classical central-arrangement count [9, Theorem 2]; see also Stanley [30, Chapter 2]. Repeated or dependent rows can make it strict. $\square$

Sign patterns $\mathbf{1}\{Xu \geq 0\}$ attained only on arrangement boundaries need not correspond to full-dimensional chambers, but chamber closures suffice for every ReLU atom: every point of the row span lies in the closure of some full-dimensional region, and a row whose sign changes among the incident regions satisfies $x_i^T u = 0$, leaving the gated feature unchanged.

3 Pricing and exact optimality

The chamber count of Proposition 2 is the size obstacle. This section shows that testing optimality does not require touching it: the exponential family of chamber tests collapses to one continuous problem, and together with a saturation condition on the active atoms it characterizes global optimality exactly.

For a residual $\nu \in \mathbf{R}^n$, define the *price* of a direction set: the chamber-restricted price and the global price are

$$\rho_s(\nu) = \max_{u \in K_s, \|u\|_2 \leq 1} \nu^T D_s X u, \quad \rho(\nu) = \max_{\|u\|_2 \leq 1} \nu^T (Xu)_+.$$

The support function of the signed atom set is $p(\nu) = \max\{\rho(\nu), \rho(-\nu)\}$, the *signed price*. Dual feasibility of ν is the family of *polar inequalities* $\nu^T a \leq \lambda$ for all $a \in \mathcal{A}$, equivalently $p(\nu) \leq \lambda$. In column-generation language, an atom with $\nu^T a > \lambda$ is a *violation*, and finding one is *separation*.

Theorem 3 (Pricing collapse). *For every $\nu \in \mathbf{R}^n$,*

$$\max_{s \in \mathcal{S}} \rho_s(\nu) = \rho(\nu).$$

Proof. For $u \in K_s$ with $\|u\|_2 \leq 1$ we have $D_s Xu = (Xu)_+$, so each chamber-restricted value is at most $\rho(\nu)$. Conversely every unit direction lies in the closure of an incident full-dimensional chamber and attains the same ReLU feature there. $\square$

So the continuum of directions prices exactly as the finitely many chambers do; the difficulty is not the continuum, and separation needs no enumeration.

Dual feasibility alone, however, does not certify an arbitrary feasible representation. Take $X = [1]$, $y = 2$, $\lambda = 1/2$, and represent the prediction $\hat{y} = 2$ by coefficient 2 on the atom $(X \cdot 1)_+ = 1$, so the mass is $\ell = 2$ and the cost is $0 + \lambda \cdot 2 = 1$. The residual is zero, hence both prices are zero and dual feasibility holds—yet the optimum $\hat{y} = 3/2$ has cost $7/8$. What is missing is that each *active* atom must saturate its polar inequality.

For a representation $\hat{y} = \sum_j c_j a_j$ with $c_j \geq 0$ and $a_j \in \mathcal{A}$, write $\ell = \sum_j c_j$ for its mass and $J(\hat{y}, \ell) = \frac{1}{2} \|\hat{y} - y\|_2^2 + \lambda \ell$ for its cost; for the corresponding (α, U) this equals $F(\alpha, U)$ in (2).

Theorem 4 (Full KKT conditions). *Let $\hat{y} = \sum_j c_j a_j$ with $c_j \geq 0$ and $a_j \in \mathcal{A}$, and put $\ell = \sum_j c_j$ and $\nu = y - \hat{y}$. This representation attains the optimal value of (3) (equivalently, $\hat{y}$ is optimal and $\ell = \gamma_{\mathcal{A}}(\hat{y})$) if and only if*

$$p(\nu) \leq \lambda \quad \text{and} \quad \nu^T a_j = \lambda \quad \text{for every } c_j > 0. \quad (5)$$

Equivalently, in the chamber program (4), both one-sided prices $\rho(\pm\nu)$ are at most λ and every nonzero block saturates its polar inequality. Every active normalized block is therefore a global price maximizer at value λ ; an inactive chamber may also contain a maximizer.

Proof. Assume (5). For any competing representation $(\hat{y}', \ell')$, dual feasibility gives $\nu^T \hat{y}' \leq \lambda \ell'$, while active saturation gives $\nu^T \hat{y} = \lambda \ell$. Hence

$$J(\hat{y}', \ell') - J(\hat{y}, \ell) = \frac{1}{2} \|\hat{y}' - \hat{y}\|_2^2 - \nu^T (\hat{y}' - \hat{y}) + \lambda (\ell' - \ell) \geq \frac{1}{2} \|\hat{y}' - \hat{y}\|_2^2.$$

Conversely, at a representation attaining the optimal value, adding a small amount of any signed atom shows $\nu^T a \leq \lambda$, and a two-sided scalar variation of each positive coefficient forces equality. In the chamber program, pairing each nonzero block's KKT inclusion with the block itself gives the same saturation condition, because the normal-cone term is orthogonal to the block. $\square$

Corollary 5 (Exact restricted stopping). *Let a representation be an exact minimizer of the restricted problem over a finite atom set. Then its active saturation conditions already hold, and the representation is globally optimal for (3) if and only if exact pricing of both signs gives $\rho(\nu) \leq \lambda$ and $\rho(-\nu) \leq \lambda$.*

Proof. Apply Theorem 4 to the finite dictionary of the restricted problem: exact restricted optimality yields saturation of its active atoms. Global optimality then reduces to the two remaining price inequalities. $\square$

The corollary concerns an exact restricted minimizer. It does not apply after an unresolved approximate solve, and this is exactly the premise the practical solver of §7 gives up.

Proposition 6 (Sparse optimal representation). *Problem (3) admits an optimum represented by at most n signed atoms, hence by at most n chamber blocks.*

Proof. Choose a finite minimum-mass representation of an optimal prediction (finite existence follows from Carathéodory’s theorem applied in $\mathbf{R}^n$). If more than n coefficients are positive, there is a nonzero h with $\sum_j h_j a_j = 0$. Active saturation in Theorem 4 gives $\lambda \sum_j h_j = \nu^T \sum_j h_j a_j = 0$, so moving the coefficients along h until one hits zero preserves both the prediction and the mass. Repeat. $\square$

The same argument applies verbatim to any finite restricted problem. The result is existential: it bounds neither every optimum nor the working set of an algorithm.

Corollary 7. *In Proposition 1, width $m \geq n$ is always sufficient.*

The dual. The dual of problem (3) will let us convert price bounds into objective bounds. It is

$$\begin{aligned} & \text{maximize} && D(\xi) = \xi^T y - \frac{1}{2} \|\xi\|_2^2 \\ & \text{subject to} && p(\xi) \leq \lambda, \end{aligned} \tag{6}$$

with variable $\xi \in \mathbf{R}^n$; the constraint is the pair of polar constraints $\rho(\pm\xi) \leq \lambda$. Weak duality is immediate: for any representation $(\hat{y}, \ell)$ and any feasible ξ ,

$$J(\hat{y}, \ell) \geq \frac{1}{2} \|\hat{y} - y\|_2^2 + \xi^T \hat{y} = D(\xi) + \frac{1}{2} \|\hat{y} - y + \xi\|_2^2 \geq D(\xi),$$

where the first inequality uses $\lambda \ell \geq \sum_j c_j \xi^T a_j = \xi^T \hat{y}$. A feasible ξ therefore certifies $J^* \geq D(\xi)$, where J^* denotes the optimal value. Restricting the polar constraints to a finite working set gives the restricted dual, and since maximizing D is minimizing $\|\xi - y\|_2^2/2$, the exact restricted dual solution is the Euclidean projection of y onto the restricted constraint set. This projection characterization drives the construction in Appendix A.

4 Ideal exact column generation

Corollary 5 suggests the method: solve the restricted problem, price the residual, add the violator, repeat. We first analyze the ideal version, in which every step is exact; Algorithm 1 states one *canonical round*. The restricted problem over the working set is called the *restricted master*, following column-generation usage; as noted in §2 it is an ordinary lasso, and the *empty master* (empty working set) has cost $J(0, 0)$. A signed atom is $\sigma(Xu)_+$, so a working set of signed atoms and a working set of signed directions are the same object, and we use whichever is convenient.

This is fully corrective column generation—all retained coefficients are refit every round—in the standard exchange template for a continuous dictionary [1, 20]. Let J_t be the restricted-master cost after t canonical rounds, with $J_0 = J(0, 0)$.

Theorem 8 (Exact canonical-round rate). *Suppose an optimum of (3) has a representation of mass at most τ , and let*

$$R = \max_{a \in \mathcal{A}} \|a\|_2 \leq \sigma_{\max}(X).$$

Algorithm 1 Ideal exact RCG (one canonical round)

Require: data X, y ; penalty λ ; working set W of signed atoms

- 1: **master:** solve the restricted master over W exactly, obtaining the prediction $\hat{y}_t$, mass ℓ_t , and residual $\nu_t = y - \hat{y}_t$
 - 2: **price:** compute $p_t = p(\nu_t)$ and an attaining signed atom a_t exactly
 - 3: **stop** if $p_t \leq \lambda$: the iterate is globally optimal (Corollary 5)
 - 4: **add:** $W \leftarrow W \cup \{a_t\}$, retaining the current representation
-

Starting from the empty master, canonical exact rounds satisfy

$$J_t - J^* \leq \frac{2\tau^2 R^2}{t + 4}. \quad (7)$$

If $\tau R = 0$, the empty master is already optimal.

Proof. Let $\Delta_t = J_t - J^*$. Exact restricted-master optimality gives the saturation identity $\nu_t^T \hat{y}_t = \lambda \ell_t$ (Corollary 5). Let an optimal representation $(\hat{y}^*, \ell^*)$ have mass $\ell^* \leq \tau$. Convexity of the squared loss gives

$$\Delta_t \leq \nu_t^T (\hat{y}^* - \hat{y}_t) + \lambda(\ell_t - \ell^*) = \nu_t^T \hat{y}^* - \lambda \ell^* \leq (p_t - \lambda) \ell^*. \quad (8)$$

In particular, if $\Delta_t > 0$ then $p_t > \lambda$ and $\Delta_t \leq \tau(p_t - \lambda)$. Keeping the current coefficients and adding the priced atom with the scalar step

$$\delta = \frac{p_t - \lambda}{\|a_t\|_2^2}$$

decreases the cost by exactly $(p_t - \lambda)^2 / (2\|a_t\|_2^2)$; we call this update the *one-coordinate comparator*. Exact reoptimization can only do better, so

$$\Delta_{t+1} \leq \Delta_t - \frac{\Delta_t^2}{2\tau^2 R^2}.$$

Write $C = 2\tau^2 R^2$. Full KKT at an optimum gives $\Delta_0 = J(0, 0) - J^* \leq \|\hat{y}^*\|_2^2 / 2 \leq C/4$, which matches (7) at $t = 0$; and if $\Delta_t \leq C/(t + 4)$ then $\Delta_{t+1} \leq C(t + 3)/(t + 4)^2 \leq C/(t + 5)$. $\square$

The bound involves no chamber count and no dimension; the data enter only through τ , the mass of an optimum, and R , the largest atom norm. Compare Proposition 2: enumeration-based formulations carry the chamber count in their problem size, whereas the rate (7) does not. Column generation pays for the dictionary through the pricing oracle instead, which motivates the study of pricing complexity in §5.

Practical solvers do more between rounds than add one atom: they re-aim directions, merge atoms, prune, and cap. The rate survives such moves under explicit conditions.

Proposition 9 (Optional moves preserve the rate conditionally). *Consider a counted round (one that advances the index t in Theorem 8) that begins at an exact KKT point of the represented finite master and interposes arbitrary modifications of the working set and representation. The recurrence of Theorem 8 holds for the round if: (i) exact signed pricing is*

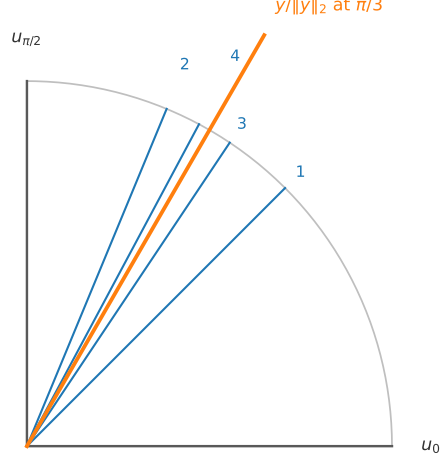


Figure 2: The nontermination instance of Theorem 10. From the atoms $u_0, u_{\pi/2}$, exact pricing adds the angular midpoint of the bracketing pair in every round (first four additions numbered). The target direction at angle $\pi/3$ is approached by bisection and never reached.

evaluated at that KKT point and returns an exact maximizer a_t ; (ii) the represented atoms and a_t are all columns of the next master, so the one-coordinate comparator remains feasible; (iii) any replacement of the representation has exact cost no larger than the comparator's; and (iv) the next counted round again begins with an exact master solve.

Proof. Conditions (i)–(ii) reproduce the inequality (8) and keep the comparator available to the next exact solve; (iii) can only improve on the comparator's decrease; (iv) restores the saturation identity used at the start of the next round. $\square$

The conditions are not vacuous: chamber compression (§7) is an exact, cost-nonincreasing replacement satisfying (ii) and (iii); the production solver nevertheless forfeits (i) and (iv) through its numerical master.

Even with everything exact, the method need not stop.

Theorem 10 (No finite termination). *There is an instance with $n = d = 2$ on which exact RCG, started from a two-atom master, adds a new atom in every round: no finite round passes the price test, although $J_t \rightarrow J^*$. Concretely, $X = I_2$, $\lambda = 1$, and $y = 4u_{\pi/3}$, where $u_\theta = (\cos \theta, \sin \theta)$, initialized with u_0 and $u_{\pi/2}$.*

On this instance the exact restricted dual is a Euclidean projection of y onto the accumulated polar constraints, and exact pricing adds the angular midpoint of the two constraints bracketing $\pi/3$ (Figure 2); since $\pi/3$ is never a dyadic midpoint of $[0, \pi/2]$, every round finds a strict violator. Appendix A gives the proof. Proposition 6 is not contradicted: the full problem admits a sparse optimal representation, while the accumulated dictionary keeps growing. Tolerance-based stopping is therefore not an implementation shortcut: by Theorem 10, even the ideal method cannot rely on exact termination.

5 The complexity of pricing

The analysis so far confines the difficulty to the pricing oracle. This section maps its complexity: pricing is a Boolean quadratic problem in disguise, which makes it NP-hard in general—but the same structure makes it boundable (§6), tractable at fixed effective rank, and coverable by sampling when the target chambers are wide.

As a computational problem: given rational X , ν , and rational c , *pricing* asks whether $p(\nu) \geq c$; the approximation version asks for a direction whose price is at least a constant fraction of $p(\nu)$.

Lemma 11 (Positive-part Boolean collapse). *For $w \in \mathbf{R}^n$ with $w \geq 0$ entrywise, with $\odot$ the componentwise product,*

$$\rho(w) = \max_{s \in \{0,1\}^n} \|X^T(s \odot w)\|_2.$$

Proof. For fixed u , maximizing $\sum_i s_i w_i x_i^T u$ over $s \in \{0,1\}^n$ selects exactly the positive entries of Xu and recovers $w^T (Xu)_+$; the finite maximum over s commutes with the maximum over the unit ball, and $\max_{\|u\|_2 \leq 1} (s \odot w)^T Xu = \|X^T(s \odot w)\|_2$. $\square$

Theorem 12 (Pricing is hard). *Pricing is NP-hard and APX-hard on rational inputs, already for $\nu = \mathbf{1}$. Consequently it admits no polynomial-time approximation scheme unless $P = NP$. The reduction does not exclude a fixed-constant approximation.*

Proof. Let $A_G \in \{-1, 0, 1\}^{|E| \times |V|}$ be an oriented incidence matrix of an unweighted graph G and set $X = A_G^T$, so $n = |V|$ and $d = |E|$. For $s \in \{0, 1\}^{|V|}$, the vector $A_G s$ has one entry per edge, equal to ± 1 exactly on edges cut by s ; hence by Lemma 11,

$$\rho(\mathbf{1})^2 = \max_{s \in \{0,1\}^{|V|}} \|A_G s\|_2^2 = \text{MaxCut}(G).$$

Since $\rho(\mathbf{1})^2$ is an integer, a rational c with $c^2 \in (k-1, k]$, computable in polynomial time, makes the question $p(\mathbf{1}) \geq c$ decide $\text{MaxCut}(G) \geq k$; the gap instances likewise admit rational thresholds. ReLU features are nonnegative, so $\rho(-\mathbf{1}) = 0$ and $p(\mathbf{1}) = \rho(\mathbf{1})$. The construction is rational. A direction whose price is within a factor $1/(1+\delta)$ of optimal induces, through its positive gate, a cut within $1/(1+\delta)^2$ of the maximum cut, so MaxCut gap hardness [19, Theorem 8.2] gives NP-hardness, APX-hardness, and the no-PTAS consequence. Conversely, a cut containing at least half the edges always exists and yields a direction with price at least $\rho(\mathbf{1})/\sqrt{2}$, so the squared transfer cannot rule out every fixed constant. $\square$

The hard instances have $d = |E|$ growing with the graph; hardness is a large-rank phenomenon. At fixed effective rank the whole problem—not just pricing—is tractable to prescribed accuracy.

Theorem 13 (Fixed effective rank). *Suppose X, y, λ are rationally encoded, $\lambda > 0$, and $r = \text{rank } X$ is fixed. Enumerating all full-dimensional chamber closures gives a rational second-order cone program (SOCP) of size polynomial in the input. For rational $\epsilon > 0$, one can compute in time polynomial in the input length and $\log(1/\epsilon)$ a point violating each conic constraint by at most ϵ whose objective is within ϵ of optimal, in the standard weak conic model.*

Proof. Proposition 2 bounds the number of closures polynomially for fixed r , and arrangements in fixed dimension are enumerable in polynomial time [11]. The sign inequalities, norm epigraphs, and a rotated-cone epigraph for the squared loss form a rational SOCP [6, §4.4.2]; the zero point bounds the total block mass by $\|y\|_2^2 / (2\lambda)$, so the relevant level set is bounded, and weak conic optimization gives the stated accuracy [25, Chapters 4 and 6]. $\square$

This is full enumeration at prescribed accuracy; it neither returns exact irrational optima in the bit model nor bounds the number of RCG rounds.

The remaining positive result concerns the cheapest oracle of all: random directions. Selected-pattern solvers [21, 23] rely on sampled gates, and multistart search needs seeds; both are governed by how much of the sphere the relevant chambers occupy. For a gate s let C_s denote its chamber and $\mu(s) = \Pr\{g \in C_s\}$ its Gaussian measure, with $g \sim \mathcal{N}(0, I_r)$ in the row span. Say a finite set T of chambers is *covered* by sampled directions when each chamber in T contains at least one of them.

Proposition 14 (Coverage of a fixed target set). *For N independent Gaussian directions and a finite target set T ,*

$$\Pr\{T \text{ is covered}\} \geq 1 - \sum_{s \in T} (1 - \mu(s))^N,$$

so with $\mu_{\min} = \min_{s \in T} \mu(s)$, a budget $N \geq \mu_{\min}^{-1} \log(|T|/\delta)$ covers T with probability at least $1 - \delta$. If $r \geq 2$ and a chamber C_s contains a center $\bar{u}$ with normalized margin

$$m_0 = \min_{i: \|x_i\|_2 > 0} \frac{|x_i^T \bar{u}|}{\|x_i\|_2 \|\bar{u}\|_2} > 0,$$

then

$$\mu(s) \geq \frac{m_0^{r-1}}{\pi(r-1)}. \quad (9)$$

Proof. The union bound gives the coverage statement. The angular distance from $\bar{u}$ to every nonzero row hyperplane is at least $\arcsin m_0$, so the open spherical cap of that radius stays inside the chamber; integrating its surface measure and using concavity of sine gives (9). The case $r = 1$ is elementary on the two-point sphere. $\square$

The direction of this result matters. A lower bound on chamber mass gives an upper bound on a sufficient sampling budget; it cannot prove that a chamber is small or that exponentially many samples are *necessary*. (If every nonzero row equals e_1^T , the positive chamber has mass 1/2 in every dimension.) What sampling actually does to the fitting objective, as measured margins shrink with dimension, is an empirical question we take up in §8.2.

6 Computable price bounds

Exact pricing is hard, but a valid *upper bound* on the price is still useful twice over: it can pass the stopping test $p(\nu) \leq \lambda$ when search cannot, and through the dual (6) it scales

the residual into a feasible dual point and hence an objective lower bound. This section constructs such bounds from the Boolean collapse, in exact arithmetic, and develops a $d \times d$ dual so that no $n \times n$ semidefinite object is ever formed: the n -sized operations are two batched matrix products, and one evaluation costs $O(nd^2)$.

6.1 Two bounds from the Boolean collapse

Fix ν and put $w = \nu_+ = \max(\nu, 0)$. Negative residual entries only lower the one-sided score, so $\rho(\nu) \leq \rho(w)$. Define

$$b_0 = X^T w, \quad \widehat{B} = \begin{bmatrix} b_0^T \\ \mathbf{diag}(w)X \end{bmatrix} \in \mathbf{R}^{(n+1) \times d},$$

with rows indexed $0, 1, \dots, n$. Substituting $s = (\mathbf{1} + \varepsilon)/2$ in Lemma 11 and using the sign symmetry $z \mapsto -z$,

$$\rho(w) = \frac{1}{2} \sqrt{\max_{z \in \{-1, 1\}^{n+1}} z^T \widehat{B} \widehat{B}^T z} \leq \bar{\rho}_{\text{joint}}(\nu) := \frac{1}{2} \sqrt{\text{SDP}(\widehat{B} \widehat{B}^T)},$$

where the first equality is exact and

$$\text{SDP}(G) = \max\{\langle G, Y \rangle \mid Y \succeq 0, \mathbf{diag}(Y) = \mathbf{1}\}$$

is the standard semidefinite relaxation of Boolean quadratic maximization. Nesterov's bound [24] gives $\bar{\rho}_{\text{joint}}(\nu) \leq \sqrt{\pi/2} \rho(\nu_+)$: past the sign truncation $\rho(\nu) \leq \rho(\nu_+)$, the joint form is within the absolute constant $\sqrt{\pi/2} \approx 1.25$ of what it bounds.

A second bound follows from the split $(t)_+ = (t + |t|)/2$: since $\nu^T (Xu)_+ \leq w^T (Xu)_+ = \frac{1}{2} w^T Xu + \frac{1}{2} \sum_i w_i |x_i^T u|$, bounding the linear term by Cauchy–Schwarz and relaxing the sign maximization in the second term by the same SDP gives

$$\bar{\rho}_{\text{split}}(\nu) = \frac{1}{2} \|X^T \nu_+\|_2 + \frac{1}{2} \sqrt{\text{SDP}(\mathbf{diag}(\nu_+) X X^T \mathbf{diag}(\nu_+))}.$$

The split form discards the coupling between the linear and Boolean parts and carries no constant-factor comparison; the joint form loses only the sign truncation beyond the SDP relaxation. Both are valid for their sign, and both remain valid when $\text{SDP}(\cdot)$ is replaced by any upper bound on it—which is what the next subsection supplies.

6.2 A small-cone dual

The relaxation $\text{SDP}(BB^T)$ for $B \in \mathbf{R}^{q \times d}$ has a $q \times q$ variable, unusable at $q \approx n$. Its diagonal dual is classical [10]; the identity below re-expresses that dual over the $d \times d$ cone, which we call the *small-cone form*.

Theorem 15 (Small-cone dual identity). *Let $B \in \mathbf{R}^{q \times d}$ have rows b_i^T . In exact arithmetic,*

$$\text{SDP}(BB^T) = \min_{\eta \in \mathbf{R}^q} \{\mathbf{1}^T \eta \mid \mathbf{diag}(\eta) \succeq BB^T\} \tag{10}$$

$$= \max_{Z \succeq 0} \left(2 \sum_i \sqrt{b_i^T Z b_i} - \text{tr } Z \right). \tag{11}$$

Zero rows of B contribute zero terms to the sum in (11) and may be deleted; after deleting them, every feasible η in (10) is entrywise positive and the matrix constraint is equivalent to $B^T \mathbf{diag}(\eta)^{-1} B \preceq I$.

Proof. The primal SDP has the strictly feasible point $Y = I$, so strong duality holds and the dual minimum in (10) is attained. Delete zero rows; congruence and equality of nonzero singular values give the inverse-form constraint. Dualizing that constraint with multiplier $Z \succeq 0$ separates over coordinates, and for each row

$$\inf_{t>0} \left(t + \frac{b_i^T Z b_i}{t} \right) = 2\sqrt{b_i^T Z b_i}.$$

A large positive multiple of $\mathbf{1}$ is strictly feasible in (10), so the exchange of min and max is exact, giving (11). $\square$

The right side of (11) is a concave function of the $d \times d$ variable Z , maximized by a few projected gradient steps whose n -sized work is the two batched products $\mathbf{diag}(BZB^T)$ and $B^T \mathbf{diag}(\cdot)B$. The essential feature is that an *inexact* Z still yields a valid bound:

Proposition 16 (Recovering a diagonal bound). *Delete zero rows of B . Choose any positive weights $\bar{c}_i > 0$ and form*

$$M = \sum_i \frac{b_i b_i^T}{\bar{c}_i}.$$

If $\bar{\lambda}_M$ is an exact or certified upper bound on $\lambda_{\max}(M)$, then $\eta_i = \bar{\lambda}_M \bar{c}_i$ is feasible in (10), so

$$\text{SDP}(BB^T) \leq \bar{\lambda}_M \sum_i \bar{c}_i. \quad (12)$$

Proof. With $\eta_i = \bar{\lambda}_M \bar{c}_i$, the inverse-form constraint of Theorem 15 reads $B^T \mathbf{diag}(\eta)^{-1} B = M/\bar{\lambda}_M \preceq I$, which holds by the definition of $\bar{\lambda}_M$. $\square$

In practice one takes $\bar{c}_i$ near $\sqrt{b_i^T Z b_i}$ for an approximately optimal Z , floored away from zero; the floor is sound only if the floored values are the ones used to form M .

From price bounds to a dual bound. Evaluate the bounds separately per sign and take, within each sign, the minimum of the valid forms:

$$\bar{\rho}_+ = \min\{\bar{\rho}_{\text{joint}}(\nu), \bar{\rho}_{\text{split}}(\nu)\}, \quad \bar{\rho}_- = \min\{\bar{\rho}_{\text{joint}}(-\nu), \bar{\rho}_{\text{split}}(-\nu)\}, \quad \bar{p} = \max\{\bar{\rho}_+, \bar{\rho}_-\} \geq p(\nu),$$

in exact arithmetic, with $\text{SDP}(\cdot)$ replaced throughout by the recovered bound (12). Maximizing the dual objective D of (6) along the ray $\xi = \theta\nu$ over the certified interval $0 \leq \theta \leq \lambda/\bar{p}$ gives

$$\theta = \begin{cases} 0, & \nu = 0, \\ \max\{\nu^T y / \|\nu\|_2^2, 0\}, & \nu \neq 0, \bar{p} = 0, \\ \max\{\min(\nu^T y / \|\nu\|_2^2, \lambda/\bar{p}), 0\}, & \nu \neq 0, \bar{p} > 0, \end{cases}$$

and $D(\theta\nu)$ is then a lower bound on J^* . A valid price bound thus becomes a computable optimality gap for any feasible model.

Remark 17 (Floating-point status). Equations (10)–(12) are exact algebra. Our implementation accumulates M in round-to-nearest floating point and calls a standard symmetric eigensolver, and that computed value can lie *below* the exact $\lambda_{\max}(M)$: with scalar rows $b_1 = 1$, $b_2 = 2^{-27}$ and $\bar{c}_1 = \bar{c}_2 = 1$, the exact value $1 + 2^{-54}$ rounds to 1, and taking $\bar{\lambda}_M = 1$ makes $\mathbf{diag}(\eta) - BB^T$ indefinite. Inflating the final scalar by one unit in the last place would cover this example but not errors accumulated earlier in the products and the eigensolve. A rigorous bound requires *outward* (directed) evaluation—each floating operation rounded in the direction that provably preserves the inequality, as in interval arithmetic—applied to the accumulation of M , the spectral bound, and the evaluation of the primal and dual objectives, and it should fail closed: report failure rather than an unverified bound. Until that path exists, the gap fields our solver reports are numerical estimates, and we flag them as such wherever they appear.

6.3 What semidefinite rounding does and does not give

The same relaxation, rounded, proposes directions. Let $B = \mathbf{diag}(|\nu|)X$, let $G = BB^T$, and define the symmetrized objective $h(u) = \sum_i |\nu_i| |x_i^T u|$ with maximum h^* over the unit ball. Suppose $Y \succeq 0$, $\mathbf{diag}(Y) = \mathbf{1}$, and $\langle G, Y \rangle \geq \kappa \text{SDP}(G)$ for a proved $0 < \kappa \leq 1$. Choose Gram vectors v_i with $Y_{ij} = v_i^T v_j$, draw a standard Gaussian g , set $\varepsilon_i = \mathbf{sign}(v_i^T g)$, and let $u_\varepsilon = B^T \varepsilon / \|B^T \varepsilon\|_2$ (zero if $B^T \varepsilon = 0$).

Proposition 18 (Symmetrized rounding). *Assume $h^* > 0$ and let $a_\kappa = 2\kappa/\pi$. Then $\mathbf{E} h(u_\varepsilon)^2 \geq a_\kappa (h^*)^2$, and for every $0 \leq \zeta < a_\kappa$, one draw satisfies*

$$\Pr\{h(u_\varepsilon) \geq \sqrt{\zeta} h^*\} \geq \frac{a_\kappa - \zeta}{1 - \zeta}.$$

For an exact SDP optimizer ($\kappa = 1$), the choice $\zeta = 1/\pi$ gives factor $1/\sqrt{\pi}$ with probability at least $1/(\pi - 1)$; independent draws amplify this threshold.

Proof. The hyperplane-rounding identity gives $\mathbf{E} \varepsilon^T G \varepsilon \geq (2/\pi) \langle G, Y \rangle$ [18], and pointwise $h(u_\varepsilon) \geq \|B^T \varepsilon\|_2$, so $\mathbf{E} h(u_\varepsilon)^2 \geq (2/\pi) \langle G, Y \rangle \geq a_\kappa (h^*)^2$, using $(h^*)^2 = \max_\varepsilon \varepsilon^T G \varepsilon \leq \text{SDP}(G)$. Normalizing by $(h^*)^2$ yields a variable $Q \in [0, 1]$ with $\mathbf{E} Q \geq a_\kappa$; if $q_0 = \Pr\{Q \geq \zeta\}$ then $a_\kappa \leq \zeta(1 - q_0) + q_0$, the tail bound. $\square$

This proposition controls h , not mixed-sign ReLU pricing, and the gap between the two can be arbitrarily large. With rows e_1, e_1, e_2 and residual $(L, -L, 1)$, every exact SDP optimizer has $Y_{12} = 1$, so rounding gives $\varepsilon_1 = \varepsilon_2$ almost surely; then $B^T \varepsilon = (\pm 2L, \pm 1)$, and even checking both orientations, the best ReLU price obtained is $1/\sqrt{4L^2 + 1}$ while $p(\nu) = 1$: the ratio tends to zero. There is no uniform conversion from symmetrized to signed pricing. Our solver therefore uses rounding only to *propose* directions and always evaluates their actual ReLU scores.

Observed tightness. Two exploratory observations (floating point, single runs) locate these bounds in practice. On ten small random instances where the price could be computed exactly by enumeration, the joint small-cone estimate exceeded the exact price by factors

Algorithm 2 Production RCG (one round)

Require: data X, y ; penalty λ ; working set; sketch rows S ; per-sign append budget n_{add} (default 8); column cap

- 1: **master:** solve the atom lasso numerically (warm active set, cached Gram columns, ridge, tolerances); gather the residual ν
 - 2: **consolidate:** solve the gated proposal (13) on a sketch; refit the projected directions on all rows; **accept only if** the full-data cost does not increase
 - 3: **price:** run sketched multistart ascent on $\pm\nu$; rescore survivors on all rows; retain orientation-distinct finalists per sign
 - 4: **add:** append up to $2n_{\text{add}}$ violators ($|a^T\nu| > \lambda$), deduplicated with orientation; refit; prune tiny coefficients; compress same-chamber groups; cap the working set
 - 5: **diagnose:** update the price-based and bound-based gap estimates; stop on any rule in the stopping paragraph below
-

1.4–2.5, where an earlier additive variant of the bound (using $|\nu|$ in place of ν_+) read 2.3–7.4, and every evaluation was numerically valid against the enumerated truth. On adversarial Max-Cut pricing instances ($|V| = 8\text{--}16$, eight graphs per size), hyperplane-rounded proposals recovered the enumeration-exact optimum in every trial (ratio 1.000), while gradient ascent from random starts, with the activation mask held fixed within each step as in §7, averaged 0.974–0.995 of it.

7 The RCG solver

The implemented solver keeps the objective and gives up the exact premises. Heuristics generate candidate directions; a candidate is accepted only when the full-data objective of problem (2) accepts it, and every returned model is feasible for that problem. The ideal method’s premises degrade as follows. The exact master becomes a floating active-set lasso with a small ridge term, tolerances, and an iteration cap. Exact pricing becomes a sketched multistart ascent whose survivors are rescored on all rows. The optional moves of Proposition 9 become gated consolidation, pruning, capping, and orientation-sensitive deduplication, none of which retains the proposition’s premises—so the rate (7) and the stopping guarantee of Corollary 5 are statements about the ideal method, not this solver. What survives is feasibility, the unchanged objective, and an independently recomputed cost for every returned model. Algorithm 2 shows one production round; the paragraphs below define its steps.

Master. Over the working set $\{u_1, \dots, u_P\}$ the master is the lasso on the feature matrix $A = [a(u_1) \ \cdots \ a(u_P)]$, whose columns are the working set’s atoms. Atoms drawn from nearby directions are highly correlated, which motivates a warm-started primal active-set method [27]: on the active set the KKT system is a small dense solve, sign-infeasible coordinates are dropped, and the most violating inactive atom enters, with a small ridge for conditioning and tolerances and an iteration cap for termination. Gram columns are computed once, in float64, when an atom first enters; the only n -sized master operation per step

is one correlation product $A^T \nu$.

Gated consolidation. Adding one atom at a time can stall in a way no single-atom move repairs: several active atoms straddle a better direction, and re-aiming any one of them alone increases the cost. The escape move is joint. Freeze each active atom’s activation pattern on a sketch—its *gate* $s_j = \mathbf{1}\{X_S u_j \geq 0\}$, where S is a uniformly drawn row subset and X_S, y_S denote the corresponding rows of X and y , with the penalty rescaled by $|S|/n$ —and solve the group lasso

$$\text{minimize } \frac{1}{2} \left\| \sum_{j=1}^P s_j \odot (X_S z_j) - y_S \right\|_2^2 + \frac{|S|}{n} \lambda \sum_{j=1}^P \|z_j\|_2, \quad (13)$$

with variables the blocks $z_1, \dots, z_P$, by block coordinate descent. The minimizer transports mass between atoms, deletes redundant blocks through group sparsity, and re-aims the survivors jointly. Because the gates are frozen, (13) is *not* the fitting objective; it is only a proposal. Each nonzero block is normalized to a direction, the true ReLU atoms are rebuilt, and the refit on all rows replaces the working set only if the computed full-data cost does not increase. We call this whole step *gated consolidation*; its cost appears as a named phase in §8.

Pricing search. Production pricing is a sketched multistart ascent. Its sketch is an importance sample: rows are drawn with probability proportional to $|\nu_i| \|x_i\|_2$ and reweighted so the sketched correlation is an unbiased estimate of $\nu^T (Xu)_+$. Seeds combine the normalized residual correlation $X^T \nu$, leading singular directions of $\text{diag}(\nu)X$ (spectral seeds), the active atoms and perturbations of them, surviving directions from previous rounds, and random draws. From each seed, projected gradient ascent maximizes the sketched estimate over the unit ball with the activation mask frozen within each step; writing g for the mask-frozen sketch gradient and $\omega > 0$ for a fixed damping coefficient, one iteration either scores both the normalized gradient and the damped step $(u + \omega g) / \|u + \omega g\|_2$ (the *two-trial* update) or only the damped step (the *damped* variant). Both residual signs are searched together. The highest-scoring orientation-distinct survivors per sign—the *finalists*, where u and $-u$ are never identified (see working-set maintenance below)—are rescored on all rows, and only full-data scores decide what is appended.

Candidate ordering. Raw signed correlation $|a^T \nu|$ remains the violation test. The exact one-coordinate decrease of appending an atom against a fixed residual,

$$\Delta(a) = \begin{cases} \frac{(|a^T \nu| - \lambda)_+^2}{2 \|a\|_2^2}, & a \neq 0, \\ 0, & a = 0, \end{cases} \quad (14)$$

can instead rank the candidates that pass it. Ranking is all it does—the violation quantity is unchanged—and correlation ordering remains the default because predicted-decrease ordering did not earn its extra cost in the controlled comparison of §8.5.

Working-set maintenance. Accepted atoms are refit, tiny coefficients are pruned, and the working set is capped; capping is applied only after a newly appended atom has participated in a refit, and pruning is followed by another refit, so the solver never prices a stale residual. Two operations deserve definitions. *Orientation-sensitive deduplication:* ReLU is not odd, so u and $-u$ are different atoms—for $X = (1, -1)^T$ the directions $u = 1$ and $u = -1$ have absolute cosine one yet produce the orthogonal features $(1, 0)^T$ and $(0, 1)^T$ —and deduplication therefore never identifies u with $-u$. *Chamber compression:* active atoms sharing a chamber and sign are merged by the conic combination $v = \sum_{j \in \mathcal{G}} |\alpha_j| u_j$, which leaves the prediction unchanged and can only decrease the penalty ($\|v\|_2 \leq \sum_{j \in \mathcal{G}} |\alpha_j|$ since all $u_j \in K_s$); this is an exact improvement, applied when eligible groups appear.

Precision. Feature-side quantities (atom builds, screening, candidate rescoring) involve no cancellation and run in float32, where accelerators deliver their throughput. The master’s KKT correlations $a_j^T \nu$ cancel catastrophically as the residual shrinks, so the master, the residual accumulation, and the small spectral factorizations run in float64. Returned objectives are recomputed in float64.

Sharding. The engine row-shards X and the residual across CPU blocks or CUDA/ROCm devices. Atom construction leaves full-height features on their shards; screening, Gram construction, rescoring, spectral seeds, and bound evaluation end in host reductions whose sizes depend on P , the candidate count, or d —not on n . The design is shard-ready rather than communication-optimal: each round still gathers an $O(n)$ residual to the host, and chamber handling can move $O(nP)$ masks, so communication grows with the sample count. Physical multi-CPU and multi-GPU speedups have not been measured; §8.5 reports the one placement experiment we have.

Stopping and reported diagnostics. The solver stops when any of the following holds: the price-based gap estimate falls below a relative threshold; the search finds no violator (after one retry with escalated search budgets); the objective decrease over a patience window of six rounds is below 10^{-7} relative while the best found price is within 5% of λ (plateau); every finalist deduplicates against the working set (no new distinct column); or a time budget or round cap is reached. Tolerance-based stopping is forced, not chosen: Theorem 10 shows even exact RCG need not terminate. The returned model always carries an independently recomputed full-data objective, a valid upper bound on J^* . Two gap estimates accompany it: a *bound-based gap estimate*, the floating evaluation of the bound-and-dual construction of §6 (an estimate for the reasons of Remark 17), and a *price-based gap estimate* that substitutes the best found price—a lower bound on $p(\nu)$ —where a valid upper bound belongs, and can therefore understate the true gap. Neither certifies optimality, and the solver labels both accordingly.

Defaults. Unless stated otherwise the experiments use the solver’s fixed defaults—no per-instance tuning: at most 100 rounds, up to 16 appended atoms per round across the two signs, a working-set cap of 512 columns, gated consolidation every round with at most 100 block passes, and sketches of 32,768 rows once n exceeds 65,536 (full data below that): an

importance sample for pricing, and a uniform subsample for gated consolidation that grows up to $4\times$ after rejected proposals.

8 Numerical experiments

Our evidence comes in two grades, and we label every result with its grade. *Controlled* experiments follow a preregistered protocol: one or two warm-up runs, three or more measured repeats per variant in balanced order (a cyclic Latin schedule, so each variant occupies each position within one count), synchronized accelerator timing, process-level page-fault and swap checks, and *objective replay*—the returned (U, α) re-evaluated on the full data by an independent code path. Variants are compared at matched quality: a candidate’s replayed objective must be within $10^{-4} \max\{1, |F_{\text{baseline}}|\}$ of the baseline’s (the *quality envelope*). Because these comparisons set the solver’s defaults, decision rules were fixed before measuring: a variant replaces the default only on a 20% median end-to-end reduction, and a placement change must not regress runtime by more than 10%. *Exploratory* results are single unrepeated runs, with objectives computed by the solver or driver as described and wall-clock measured in an unrecorded machine power state; we use them for orders of magnitude and cross-method comparisons of objective values, not for fine timing claims. The accompanying result records include the configurations, raw measurements, and run metadata. The hardware throughout is one laptop: an AMD Ryzen 7 7735U (eight cores, sixteen threads, OpenBLAS and OpenMP capped at eight threads) with an integrated Radeon 680M GPU reached through Torch/ROCm.

8.1 Planted instances (exploratory)

We first ask whether the solver reaches good objective values at all, on instances where a strong reference value is available. A *planted* instance draws X with i.i.d. standard normal entries and sets y as the noiseless output of k fixed teacher atoms; the *teacher reference* is the lasso solved over the teacher’s own atoms, a feasible upper bound on J^* (not a proven optimum).

Across five planted configurations spanning $n = 500$ to 5000 , $d = 12$ to 32 , and $k = 3$ to 8 , the solver’s returned objective exceeded the teacher reference by between 6×10^{-6} and 9.6×10^{-5} relative, in 0.3 to 9.3 seconds, with the search still pricing about 4% above λ at exit. For calibration, a single-column refinement baseline (per-atom refinement, no gated consolidation) finished 2% to 56% above the reference on four of the five instances (it was not run on the fifth) while taking $5\text{--}15\times$ longer—the gap that motivated the gated step. On one instance small enough to enumerate all 818 patterns ($n = 30$, $d = 3$), the heuristic search returned the enumeration-exact price in all ten rounds, and the final exact price exceeded λ by less than 10^{-8} : consistent with global optimality up to master tolerances, on that instance only.

Against tuned baselines, on a planted $n = 10^5$, $d = 16$, $k = 3$ instance: RCG reached objective 0.31408 (test MSE 1.3×10^{-10} , seven active atoms) in 42 seconds; Adam, AdamW, and SGD—width 64, each grid point of three learning rates $\times$ two seeds given the *full* 42-second budget, roughly six times RCG’s total compute per optimizer—ended at best

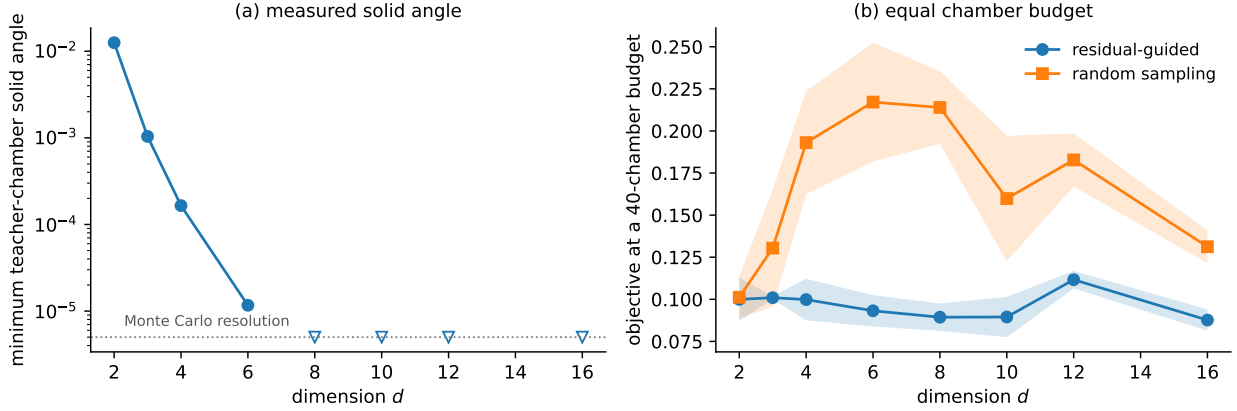


Figure 3: Sampling and dimension, planted $n = 50$, $k = 2$ (exploratory; means and standard deviations over three repeats). (a) Measured minimum teacher-chamber solid angle; open markers indicate values below the Monte Carlo resolution of 5×10^{-6} . (b) Objective at a fixed 40-chamber budget: random gates versus residual-guided selection through the same exact master.

objectives of 2018, 1672, and 1966 respectively (best test MSE 0.035), with grid medians 4–10 $\times$ their bests. On this teacher-realizable instance the first-order methods do not approach the convex optimum at any grid point.

8.2 Sampling and dimension (exploratory)

A natural question is whether residual-guided pricing offers a real advantage over random sampling of patterns. Proposition 14 says sampling covers wide chambers cheaply; the empirical question is what happens to the fitting objective as the relevant chambers narrow. On planted instances with $n = 50$, $k = 2$, and d from 2 to 16 (three repeats each), the measured minimum teacher-chamber solid angle falls from 1.3×10^{-2} at $d = 2$ to below the 5×10^{-6} Monte Carlo resolution by $d = 8$ (Figure 3a). At a fixed budget of 40 chambers, both selection rules use the same exact lasso master and differ only in where the chambers come from: random Gaussian gates versus residual-guided search. Sampling’s objective degrades by up to 2.4 $\times$ as d grows while the residual-guided objective stays flat (Figure 3b). This is the practical face of the collapse theorem: priced atoms are placed by the residual, and random ones rarely land in the active chambers once those are narrow. These runs used the reference chamber implementation; the comparison isolates column *selection*. A direct comparison with selected-pattern solvers would confound selection with master implementations and stopping rules, so we leave it to the next-steps list in §9.

8.3 Real data (exploratory)

We now compare the solver with tuned first-order baselines on real data at budget-matched wall clock. Three datasets: California housing ($15,480 \times 9$ after split), Covtype ($435,759 \times 55$, least-squares classification of one class), and YearPredictionMSD ($386,509 \times 91$). Protocol: a 75/25 train/test split (seed 0), features and target standardized by training statistics,

Dataset	Method	Objective	vs. RCG	Test MSE	Runs	Time
California $n = 15,480, d = 9$	RCG (32 atoms)	2457	—	0.263	1	17 s
	Adam best / median	4445 / 8433	+81% / +243%	0.273	9	20 s total
	AdamW best / median	4936 / 7845	+101% / +219%	0.396	15	33 s total
	SGD best / median	9628 / 15337	+292% / +524%	0.509	12	20 s total
Covtype $n = 435,759, d = 55$	RCG (85 atoms)	133911	—	0.563	1	1137 s
	Adam best / median	126528 / 129229	−5.5% / −3.5%	0.512	9	1137 s total
	AdamW best / median	130501 / 140806	−2.5% / +5.1%	0.530	15	1896 s total
	SGD best / median	152790 / 236315	+14% / +76%	0.562	6	1137 s total
MSD $n = 386,509, d = 91$	RCG (101 atoms)	134741	—	0.669	1	1959 s
	Adam best / median	130105 / 130250	−3.4% / −3.3%	0.644	9	1959 s total
	AdamW best / median	130963 / 134208	−2.8% / −0.4%	0.647	15	3265 s total
	SGD best / median	145757 / 218601	+8.2% / +62%	0.669	6	1959 s total

Table 1: Real data at budget-matched wall clock (exploratory; single runs). All methods minimize the same objective; “vs. RCG” is the signed relative difference of the method’s objective from RCG’s. Test MSE is in target-variance units, best run shown for baselines. “Total” times are tuning-grid totals; see the protocol text for per-run budgets and the AdamW exception.

bias column appended; λ set to 10^{-2} times a sketch estimate of the smallest λ with empty solution; every method minimizes the same objective, with baselines scored on the weight-decay form (1) recomputed from their final weights in float64. RCG ran with a 30-round cap, its one non-default setting for these runs, and the matched budget T below is its wall clock under that cap. Baselines are width-128 networks, minibatched at large n . The tuning grids are budget-matched to T : Adam (three learning rates $\times$ three seeds) and SGD (five learning rates $\times$ three seeds) each receive T in total, split evenly per run; AdamW’s decoupled decay rate needs its own sweep (five values at Adam’s per-run budget), so its grid total is $1.67T$. On California the grid totals are floored at 20 seconds against $T = 17$, an overshoot that favors the baselines. Divergent runs (non-finite objectives) are dropped from best and median statistics; SGD kept 12/15, 6/15, and 6/15 runs on the three datasets. Three further asymmetries, disclosed: an unbalanced network’s weight-decay cost exceeds the atomic cost of the same predictor, so baseline objectives can be slightly overstated (exact per-neuron rebalancing would remove this, but the saved records contain objectives, not weights); RCG’s objective is the solver’s final full-data float64 value rather than the controlled protocol’s independent replay; and Adam’s best learning rate landed at the grid edge on California and MSD, so those grids may not bracket its optimum.

Table 1 and Figure 4 give the results. The outcome splits by dataset. On California, RCG’s objective is 45% below the best tuned grid point (81% of RCG’s value above it), the tuned medians are $3.2\text{--}6.3\times$ RCG’s value, and held-out MSE follows the same order; the solver’s price-based gap estimate fell below 1% of the objective at round 24. On Covtype and MSD, RCG stopped at its 30-round cap with the search still pricing atoms at $1.7\text{--}2.0 \times \lambda$ —columns remained to add—and the best tuned Adam runs finished 5.5% and 3.4% lower, with correspondingly lower test MSE. Two observations temper the losses. The tuned methods pay a tuning bill the convex method does not: SGD’s median run is 50–55% above its own best on the large datasets, and it diverges outright at learning rates large enough

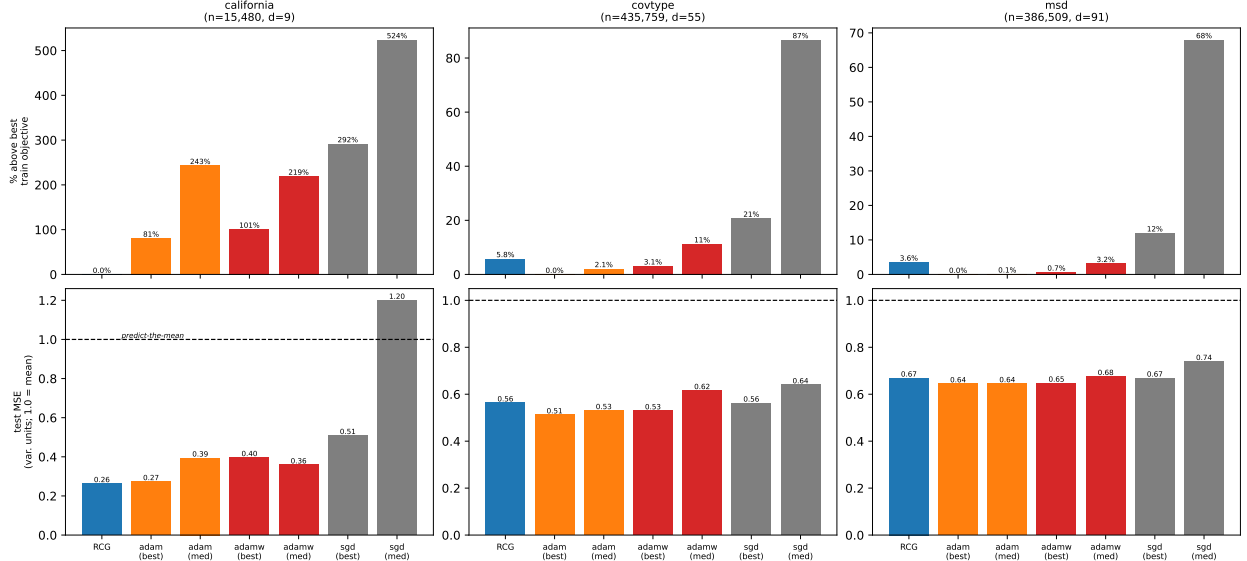


Figure 4: Real-data comparison at matched wall clock (exploratory). Top: training objective as percent above the per-dataset best. Bottom: held-out MSE in target-variance units; the dashed line is predict-the-mean.

to move fast (9 of 15 runs on each large dataset). And RCG’s exits are round-capped, not converged, so its large-dataset values are not the method’s limit. We do not attribute the split to a property of the optima (such as sparsity); with the tools of this paper that would be conjecture, and we leave it as one.

8.4 Scale (exploratory)

On planted instances at fixed $d = 16$, $k = 3$, single CPU runs under a 60-round cap: $n = 2 \times 10^4$ ran in 24.8 seconds, 10^5 in 50 seconds, 5×10^5 in 75 seconds, and 2×10^6 in 173 seconds—100 $\times$ the rows for 7 $\times$ the time—with returned objectives between $+3.9 \times 10^{-5}$ and $+1.3 \times 10^{-3}$ relative to the teacher reference (itself only a feasible upper bound). These are not convergence claims: every run at $n \geq 10^5$ reached the round cap with the search still reporting violations, increasingly so at scale (up to $6.4 \times \lambda$ at $n = 2 \times 10^6$). The sublinear growth reflects sketch-bounded search plus $O(n)$ full-data passes, consistent with the communication accounting of §7. At $n = 10^6$ a paired run of the same Torch engine put the integrated GPU at 115 seconds versus 109 on the CPU—an early hint, confirmed under protocol below, that accelerator residency is not automatically a win on this shared-memory machine.

8.5 Controlled implementation experiments

The controlled experiments ask configuration questions about the solver itself, under the protocol and decision rules stated at the top of this section. The end-to-end instance is a deterministic planted problem with $n = 10^4$, $d = 16$, three teacher atoms, $\lambda = 0.1$, seed zero.

Variant	Objective	Time (s)	IQR (s)	Ratio
two-trial, correlation (default)	0.3209115	5.447	0.069	1.000
damped, correlation	0.3209134	5.581	0.497	1.025
damped, predicted-decrease	0.3209221	4.456	0.187	0.818

Table 2: CPU variants at $n = 10^4$, $d = 16$ (controlled). Medians over three balanced-order measured runs after one warm-up; IQR is the interquartile range of the three runs by linear-interpolation quartiles, and Ratio is median time over the default’s. All objectives are independently replayed and pass the quality envelope.

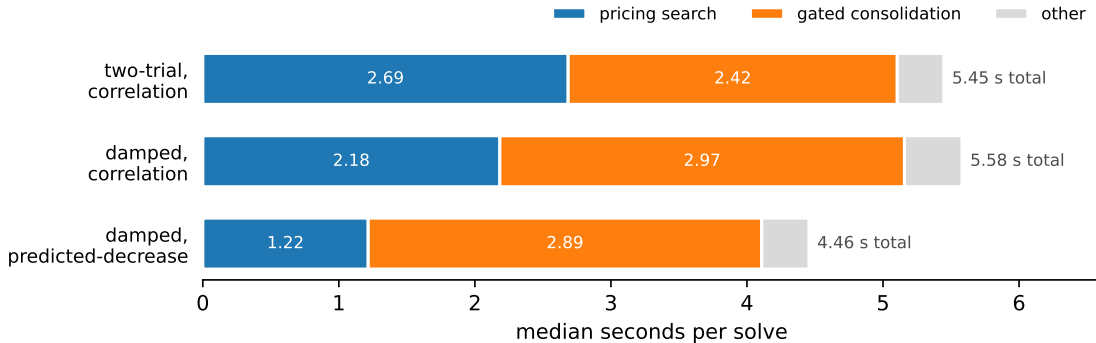


Figure 5: Where the time goes: per-phase median seconds for the three CPU variants of Table 2. Cutting pricing search exposes gated consolidation as the dominant phase.

CPU variants. Three single-host NumPy configurations: the default (two-trial ascent, correlation ordering), damped ascent alone, and damped ascent with predicted-decrease ordering. Table 2 reports medians of three measured runs after one warm-up; all replayed objectives pass the quality envelope. Damped ascent alone removes one forward product per search iteration yet is 2.5% slower end to end; combined with predicted-decrease ordering it is 18.2% faster—short of the 20% bar, so the default stands. No measured process incurred a major page fault or acquired swap, though system-wide swap activity was recorded as an interference warning during some runs. The phase decomposition (Figure 5) is the real finding: the combined variant cuts pricing search from 2.69 to 1.22 seconds, at which point gated consolidation (2.89 s) becomes the largest phase; further pricing reductions would now have limited effect on the total, and the gated phase is the natural next target.

Candidate-ordering and reuse probes. A probe that replays one component in isolation tested candidate ordering across 20 saved solver states (960 scalar checks of the identity (14) against directly computed one-coordinate updates—all passed). Predicted-decrease ordering changed the selection in only 3 of 20 states; on those reversals its median fully-refit gain was $0.59\times$ correlation’s, and computing the candidate norms it needs made rescoring $1.65\times$ slower. A second probe tested reusing the finalists’ already-built feature columns instead of rebuilding them: at $n = 1.2 \times 10^5$, $d = 64$ (eight balanced repeats after two warm-ups), reuse reproduced the same eight oriented directions and features exactly but

took $1.07\times$ the rebuild path’s time, with a 38 MB peak memory increase against a ≈ 1.8 GB ceiling. Both alternatives remain available as non-default options.

Device placement. Holding the Torch float32 engine fixed and moving only the pricing search to the integrated GPU: host search took a median 7.67 seconds (IQR 0.09), device search 11.07 seconds (IQR 1.72)—44% slower, failing the 10% regression rule, with the entire loss in the pricing phase (6.13 versus 2.30 seconds). The result concerns one shared-memory integrated GPU and says nothing about discrete or multiple GPUs; host search remains the local default.

9 Conclusions

Convexity moves the hardness of two-layer ReLU fitting into a separation oracle, and this paper analyzed that oracle in both directions. With an exact oracle, column generation carries full optimality conditions and an $O(1/t)$ rate that never sees the exponential chamber count—but provably need not terminate, so tolerance-based stopping is intrinsic to the problem, not a concession of the implementation. Without one, pricing is NP-hard and APX-hard already at the all-ones residual, yet boundable: a $d \times d$ semidefinite dual upper-bounds the price in exact arithmetic and converts any valid bound into an objective gap. In between sits the implemented solver, which accepts a heuristic proposal only when the full-data objective does, preserving feasibility and the objective while giving up the exact premises, and whose measured behavior is competitive: near-reference values on planted data, robustness to dimension at a fixed working-set budget where random sampling degrades, a 45% objective win over tuned baselines on one real dataset and 3.4–5.5% losses on two others at budget-matched wall clock, and a phase profile that names gated consolidation as the next bottleneck.

The limitations are equally concrete. The comparative evidence is exploratory: single runs, solver-computed objectives, unrecorded machine power state; the controlled protocol currently covers only implementation variants on one planted instance and one machine, with three repeats. The floating bound path is not outward-evaluated, so the solver reports gap estimates, not certificates. And the real-data losses occurred at round caps, so the convex method’s large-scale limit is unmeasured. The corresponding next steps are mechanical to state: extend the round caps on the two losing datasets, where the search was still finding violators; rerun the real-data comparison under the controlled protocol, with baselines scored on rebalanced weights; compare directly against selected-pattern solvers such as SCNN and CRONOS at matched budgets; implement the outward (directed) bound evaluation of Remark 17, fail-closed, which is one directed-rounding engineering pass; attack the gated phase; and measure discrete-GPU and multi-device execution on hardware that has it.

One mathematical question stands out. In exact arithmetic, the only structural slack in the positive-part route of §6 is the sign truncation $\rho(\nu) \leq \rho(\nu_+)$: closing it requires a relaxation that couples the gate pattern of a direction with the direction itself, and we know of no published bound that exploits this realizability structure. We leave it as the main open problem.

Code and result availability. The implementation, result records, and complete reproduction instructions are available in the [online supplement](#). The controlled records include raw measurements, data and model digests, and software and hardware metadata; the exploratory records preserve their configurations, seeds, tuning grids, and budgets.

A The nontermination construction

We prove Theorem 10. Let $X = I_2$, $\lambda = 1$, and $y = 4u_{\pi/3}$, where $u_\theta = (\cos \theta, \sin \theta)$, and initialize the master with the atoms of $u_0 = e_1$ and $u_{\pi/2} = e_2$. In the positive quadrant the price of a nonnegative residual is its Euclidean norm. By the projection characterization of the restricted dual (§3), the exact restricted dual solution is the Euclidean projection of y onto $\{\nu \mid |u_\theta^T \nu| \leq 1 \text{ for every accumulated } \theta\}$; along the constructed trajectory the lower inequalities are inactive. Initially the projection is $(1, 1)$, so exact pricing adds $u_{\pi/4}$.

Suppose adjacent accumulated angles $\varphi_1 < \pi/3 < \varphi_2$ bracket the target. Write $\varphi_m = (\varphi_1 + \varphi_2)/2$ for the midpoint and $h = (\varphi_2 - \varphi_1)/2$ for the half-width. The two tangent lines meet at $v = \sec(h) u_{\varphi_m}$. The projection normal-cone condition at v reduces to

$$4 \sin(2h/3) \geq \tan h,$$

which holds for $0 < h \leq \pi/4$ by the elementary chord bounds for sine and tangent; and v satisfies every non-adjacent accumulated constraint because $\sec(h) \cos(\theta - \varphi_m) \leq 1$ whenever $|\theta - \varphi_m| \geq h$. Thus v is the exact restricted dual solution, and exact pricing adds the midpoint angle φ_m .

The target $\pi/3$ sits at relative position $2/3$ of $[0, \pi/2]$ and remains at relative position $1/3$ or $2/3$ of the bracketing interval after every bisection; it is never a dyadic midpoint. Hence every finite round has price $\|v\|_2 = \sec h > 1 = \lambda$ and adds a new atom. The bracket shrinks, so $J_t \rightarrow J^*$, but the algorithm never stops. $\square$

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